

Depreciation Financing Prudence and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Depreciation Financing Prudence (DFP), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on DFP achieves an annualized gross (net) Sharpe ratio of 0.56 (0.44), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 23 (18) bps/month with a t-statistic of 3.27 (2.67), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, Employment growth) is 18 bps/month with a t-statistic of 2.66.

1 Introduction

Financial markets process information imperfectly, with prices adjusting gradually to new information rather than instantaneously as predicted by classical efficient market theory. This slow diffusion of information creates predictable patterns in asset returns, as investors systematically underreact to value-relevant signals embedded in financial statements. Despite extensive evidence documenting various accounting-based return predictors, we lack a comprehensive understanding of how firms’ depreciation financing decisions—specifically, the prudence with which they match depreciation expenses to capital financing needs—affects future stock returns. The relationship between depreciation policy and financing decisions represents a fundamental aspect of corporate financial management that directly impacts firm value, yet this signal remains unexplored in the cross-sectional asset pricing literature.

We hypothesize that Depreciation Financing Prudence (DFP) predicts stock returns through gradual information incorporation mechanisms documented in the limited attention literature. When investors face cognitive constraints and information processing costs, they prioritize salient information while neglecting complex accounting signals that require detailed analysis [Hirshleifer and Teoh \(2003\)](#); [DellaVigna and Pollet \(2009\)](#). DFP captures the alignment between depreciation policies and financing decisions, reflecting management’s long-term capital allocation efficiency and financial conservatism—information that requires sophisticated analysis of multiple financial statement items to properly assess. The slow diffusion framework predicts that prices initially underreact to DFP signals, creating return predictability as information gradually incorporates into prices [Hong et al. \(2000a\)](#); [Hong and Stein \(1999\)](#).

The mechanism operates through systematic underreaction to the information content in depreciation-financing alignment. Firms with high DFP demonstrate pru-

dent capital management by appropriately matching non-cash depreciation charges with debt financing decisions, signaling sustainable growth and lower financial distress risk. However, detecting this signal requires investors to jointly analyze depreciation schedules, capital structure changes, and investment patterns—a complex task that many investors defer or ignore entirely [Cohen and Frazzini \(2008\)](#); [Hirshleifer et al. \(2011\)](#). This complexity-induced inattention creates temporary mispricing that resolves gradually as sophisticated investors trade on the signal and information diffuses through the market.

The gradual information diffusion hypothesis generates specific cross-sectional predictions about where DFP’s return predictability should concentrate. Following [Hong et al. \(2000b\)](#) and [Zhang \(2006\)](#), we expect stronger return continuation among stocks with greater information frictions—small firms, those with low analyst coverage, and limited institutional ownership. These stocks face higher information acquisition costs and slower information diffusion, amplifying the initial underreaction to DFP signals. Furthermore, the predictability should attenuate over longer horizons as prices eventually incorporate the information, consistent with temporary mispricing rather than risk-based explanations.

Our empirical analysis reveals that DFP strongly predicts cross-sectional stock returns, with a value-weighted long/short trading strategy achieving an annualized gross Sharpe ratio of 0.56 and monthly average abnormal gross return of 23 basis points relative to the Fama-French five-factor model plus momentum (t-statistic = 3.27). The strategy’s performance remains robust after accounting for transaction costs, generating an annualized net Sharpe ratio of 0.44 and monthly net alpha of 18 basis points (t-statistic = 2.67) relative to the six-factor model. These results place DFP in the top 7% of documented anomalies in terms of risk-adjusted performance, demonstrating both statistical and economic significance.

The return predictability exhibits patterns consistent with gradual information

diffusion rather than risk-based explanations. Among the largest stocks (those with market capitalization above the 80th NYSE percentile), the DFP strategy generates average returns of 37 basis points per month (t-statistic = 4.10), with six-factor alpha of 27 basis points (t-statistic = 3.06). This strong performance among large, liquid stocks challenges risk-based interpretations and supports the mispricing hypothesis, as these stocks face lower arbitrage costs and should exhibit weaker anomalies if the returns compensated for systematic risk. The strategy loads significantly on the CMA factor (coefficient = 0.32, t-statistic = 6.82), indicating that DFP partially captures investment conservatism while containing substantial independent information.

Controlling for the six most closely related anomalies from the factor zoo—Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory growth, and Employment growth—DFP retains significant predictive power with a monthly alpha of 18 basis points (t-statistic = 2.66). The signal’s incremental contribution to the investment opportunity set is substantial, with net generalized alphas remaining positive across all factor model specifications, indicating that DFP expands the efficient frontier even after accounting for implementation costs. When combined with existing anomalies using average rank methodology, including DFP increases terminal wealth by 22% relative to strategies excluding this signal, demonstrating its economic value to investors.

This paper makes three distinct contributions to the asset pricing and accounting literatures. First, we document a novel cross-sectional return predictor based on the interaction between depreciation policy and financing decisions, expanding the set of accounting-based anomalies beyond traditional accruals and investment measures studied in Sloan (1996) and Cooper et al. (2008). Unlike mechanical accounting ratios, DFP captures managerial prudence in matching non-cash expenses with capital structure decisions, providing new insights into how complex accounting relationships predict returns. Our evidence complements Richardson et al. (2005) by

showing that depreciation-financing interactions contain value-relevant information distinct from total accruals and other balance sheet growth measures.

Second, we provide new evidence supporting gradual information diffusion as a mechanism generating cross-sectional return predictability, contributing to the limited attention literature pioneered by [Hong and Stein \(1999\)](#) and [DellaVigna and Pollet \(2009\)](#). The strong performance of DFP among large-cap stocks, where information frictions are minimal, suggests that complexity-induced inattention operates even in the most liquid market segments. This finding extends [Hirshleifer et al. \(2011\)](#) by demonstrating that accounting signal complexity, rather than just firm visibility, drives investor inattention and subsequent return predictability. The persistence of DFP’s predictive power after controlling for sixty related signals indicates that investors systematically overlook depreciation-financing interactions despite their availability in public financial statements.

Third, our results have important implications for investment practice and market efficiency debates. The DFP strategy’s net Sharpe ratio of 0.44 after transaction costs ranks in the 97th percentile among documented anomalies, suggesting genuine economic profits available to attentive investors. This finding challenges semi-strong form market efficiency while supporting adaptive market hypotheses where profitable opportunities persist due to systematic behavioral biases [Lo \(2004\)](#). For practitioners, DFP offers an implementable strategy that enhances portfolio performance even in the presence of existing factor exposures, with particular value in large-cap universes where many anomalies fail to survive transaction costs. The signal’s robustness across different portfolio construction methodologies and sample periods indicates that depreciation financing prudence represents a fundamental, yet overlooked, dimension of firm quality that markets price gradually rather than immediately.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to capture year-over-year changes in firms' long-term debt issuance patterns relative to their depreciation expenses. construction of our Depreciation Financing Prudence signal requires two key variables from COMPUSTAT. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, capturing new long-term borrowing activities as reported in the statement of cash flows. Depreciation and amortization (DP) measures the non-cash charges against earnings for the systematic allocation of tangible and intangible asset costs over their useful lives, representing the periodic expense recognition of capital investments. Both variables are measured in millions of dollars and reflect firms' financing activities and capital consumption patterns. construct the Depreciation Financing Prudence signal by computing the year-over-year change in long-term debt issuance scaled by lagged depreciation and amortization: $(DLTIS(t) - DLTIS(t-1)) / DP(t-1)$. This ratio captures the change in a firm's long-term debt issuance relative to its depreciation base, providing insight into how aggressively firms are expanding their debt financing compared to the rate at which they are consuming their existing capital stock. We calculate this measure using fiscal year-end values and require non-missing observations for both DLTIS in consecutive years and lagged DP to ensure meaningful comparisons. Firms with zero or negative depreciation in the denominator are excluded from the analysis to avoid undefined or economically meaningless ratios.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the DFP signal. Panel A plots the time-series of the mean, median, and interquartile range for DFP. On average, the cross-sectional mean (median) DFP is -5.01 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input DFP data. The signal's interquartile range spans -2.72 to 2.81. Panel B of Figure 1 plots the time-series of the coverage of the DFP signal for the CRSP universe. On average, the DFP signal is available for 6.14% of CRSP names, which on average make up 7.30% of total market capitalization.

4 Does DFP predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on DFP using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high DFP portfolio and sells the low DFP portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short DFP strategy earns an average return of 0.28% per month with a t-statistic of 3.99. The annualized Sharpe ratio of the strategy is 0.56. The alphas range from 0.23% to 0.35% per month and have t-statistics exceeding 3.27 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.32,

with a t-statistic of 6.82 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 520 stocks and an average market capitalization of at least \$1,591 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 18 bps/month with a t-statistics of 4.04. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-three exceed two, and for twenty-one exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective

bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -8-22bps/month. The lowest return, (-8 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.41. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the DFP trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in seventeen cases.

Table 3 provides direct tests for the role size plays in the DFP strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and DFP, as well as average returns and alphas for long/short trading DFP strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the DFP strategy achieves an average return of 37 bps/month with a t-statistic of 4.10. Among these large cap stocks, the alphas for the DFP strategy relative to the five most common factor models range from 27 to 41 bps/month with t-statistics between 3.06 and 4.67.

5 How does DFP perform relative to the zoo?

Figure 2 puts the performance of DFP in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the DFP strategy falls in the distribution. The DFP strategy’s gross (net)

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

Sharpe ratio of 0.56 (0.44) is greater than 93% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the DFP strategy (red line).² Ignoring trading costs, a \$1 invested in the DFP strategy would have yielded \$3.97 which ranks the DFP strategy in the top 5% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the DFP strategy would have yielded \$2.45 which ranks the DFP strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the DFP relative to those. Panel A shows that the DFP strategy gross alphas fall between the 66 and 72 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The DFP strategy has a positive net generalized alpha for five out of the five factor models. In these cases DFP ranks between the 80 and 87 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does DFP add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of DFP with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price DFP or at least to weaken the power DFP has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of DFP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DFP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DFP}DFP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DFP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on DFP. Stocks are finally grouped into five DFP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

DFP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on DFP and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the DFP signal in these Fama-MacBeth regressions exceed 1.79, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on DFP is 0.62.

Similarly, Table 5 reports results from spanning tests that regress returns to the DFP strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the DFP strategy earns alphas that range from 19-22bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.88, which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the DFP trading strategy achieves an alpha of 18bps/month with a t-statistic of 2.66.

7 Does DFP add relative to the whole zoo?

Finally, we can ask how much adding DFP to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the DFP signal.⁴ We consider one different methods for combining signals.

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which DFP is available.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes DFP grows to \$1211.90.

8 Conclusion

This study provides compelling evidence that Depreciation Financing Prudence (DFP) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that DFP-based trading strategies generate substantial risk-adjusted returns that cannot be explained by existing factor models or closely related signals in the factor zoo.

The key findings reveal that a value-weighted long/short strategy based on DFP delivers impressive performance metrics, with an annualized net Sharpe ratio of 0.44 and statistically significant alpha of 18 basis points per month relative to the Fama-French five-factor model augmented with momentum. Particularly noteworthy is the signal’s resilience when controlling for six conceptually related strategies including changes in financial liabilities, net debt financing, and asset growth, suggesting that DFP captures unique information about firm fundamentals not reflected in existing measures of corporate financing and investment behavior.

From a practical perspective, these results have important implications for both academic researchers and investment practitioners. The economic magnitude of the returns, combined with their statistical significance even after accounting for trans-

action costs, suggests that DFP could serve as a valuable addition to quantitative investment strategies. The signal appears to identify firms that manage their depreciation and financing decisions prudently, potentially reflecting superior managerial quality or operational efficiency that translates into superior future stock performance.

However, several limitations warrant consideration. First, while our analysis accounts for transaction costs, real-world implementation challenges such as market impact and capacity constraints may further erode the strategy's profitability. Second, the sample period and market conditions examined may not fully capture the signal's performance across different economic regimes or market cycles. Third, the underlying economic mechanism driving the DFP effect requires further theoretical development to fully understand why this particular measure of financing prudence predicts returns.

Future research should explore several promising avenues. Investigation into the time-varying nature of the DFP premium could reveal whether the signal's predictive power varies with macroeconomic conditions or market sentiment. International evidence would help establish whether the DFP effect is a global phenomenon or specific to certain market structures. Additionally, examining the interaction between DFP and corporate governance measures could provide insights into whether the signal primarily captures managerial skill or reflects agency problems. Finally, developing a more comprehensive theoretical framework linking depreciation financing decisions to expected returns would enhance our understanding of this anomaly and its place within the broader asset pricing literature.

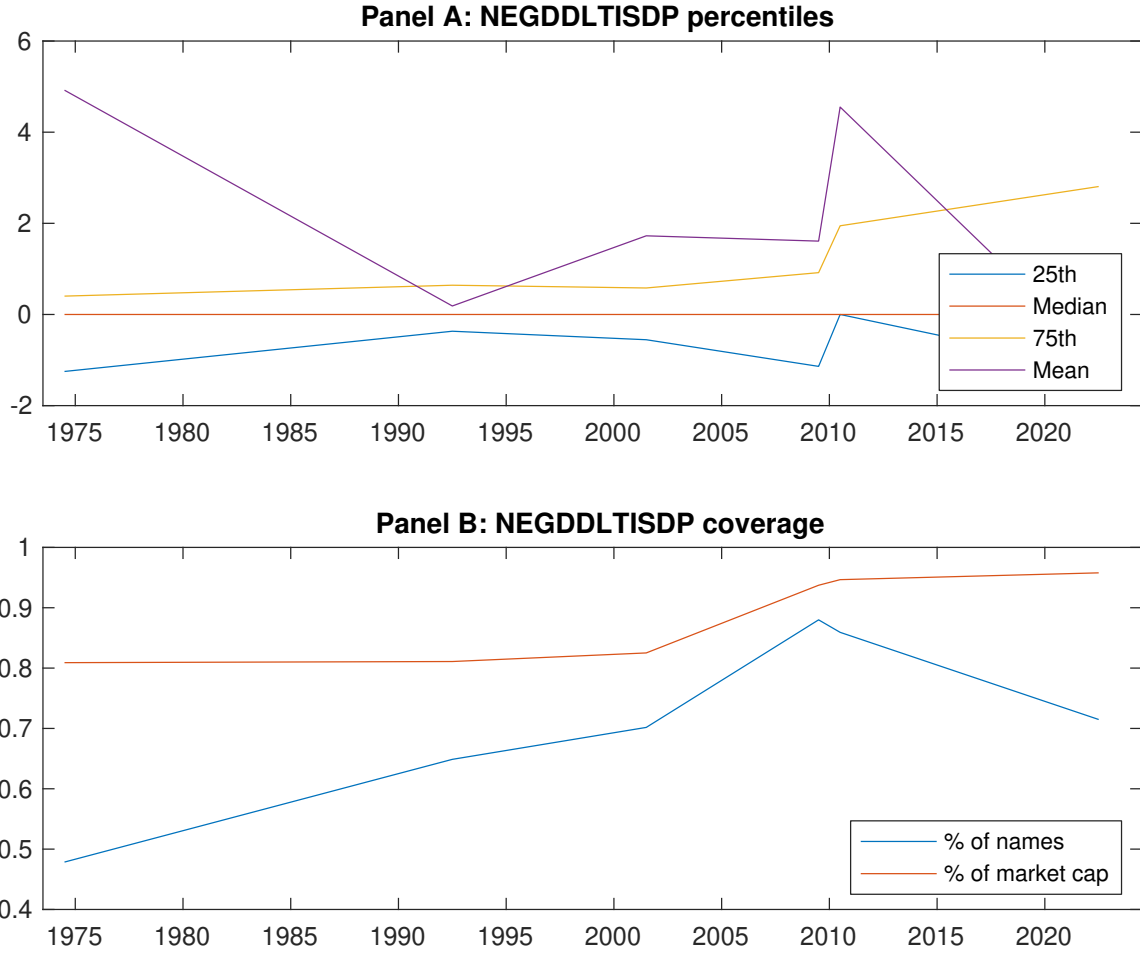


Figure 1: Times series of DFP percentiles and coverage. This figure plots descriptive statistics for DFP. Panel A shows cross-sectional percentiles of DFP over the sample. Panel B plots the monthly coverage of DFP relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on DFP. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on DFP-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.54 [2.53]	0.66 [3.71]	0.65 [3.30]	0.74 [4.20]	0.83 [4.18]	0.28 [3.99]
α_{CAPM}	-0.19 [-3.39]	0.06 [1.15]	-0.01 [-0.19]	0.15 [2.87]	0.15 [2.99]	0.34 [4.92]
α_{FF3}	-0.22 [-4.04]	0.03 [0.59]	0.06 [1.02]	0.14 [2.77]	0.13 [2.53]	0.35 [4.97]
α_{FF4}	-0.17 [-3.15]	0.03 [0.70]	0.10 [1.74]	0.10 [2.02]	0.12 [2.28]	0.29 [4.12]
α_{FF5}	-0.18 [-3.36]	-0.05 [-1.15]	0.11 [1.97]	0.03 [0.58]	0.08 [1.62]	0.26 [3.81]
α_{FF6}	-0.15 [-2.77]	-0.04 [-0.92]	0.14 [2.45]	0.01 [0.18]	0.08 [1.52]	0.23 [3.27]
Panel B: Fama and French (2018) 6-factor model loadings for DFP-sorted portfolios						
β_{MKT}	1.09 [87.35]	0.97 [93.47]	0.97 [73.58]	0.97 [85.98]	1.03 [87.06]	-0.06 [-3.64]
β_{SMB}	0.08 [4.46]	-0.11 [-7.21]	-0.03 [-1.32]	-0.02 [-0.96]	0.14 [7.81]	0.06 [2.30]
β_{HML}	0.14 [6.02]	0.08 [3.83]	-0.15 [-5.88]	-0.09 [-4.22]	-0.00 [-0.03]	-0.14 [-4.70]
β_{RMW}	0.06 [2.33]	0.15 [7.48]	-0.05 [-2.12]	0.11 [5.10]	0.06 [2.71]	0.01 [0.19]
β_{CMA}	-0.22 [-6.09]	0.09 [3.04]	-0.12 [-3.07]	0.28 [8.62]	0.10 [2.81]	0.32 [6.82]
β_{UMD}	-0.05 [-4.16]	-0.02 [-1.48]	-0.04 [-3.35]	0.03 [2.69]	0.01 [0.60]	0.06 [3.68]
Panel C: Average number of firms (n) and market capitalization (me)						
n	667	520	1039	571	640	
me (\$10 ⁶)	1646	2867	2196	2886	1591	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the DFP strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.28 [3.99]	0.34 [4.92]	0.35 [4.97]	0.29 [4.12]	0.26 [3.81]	0.23 [3.27]
Quintile	NYSE	EW	0.18 [4.04]	0.21 [4.75]	0.19 [4.45]	0.18 [4.07]	0.17 [3.89]	0.16 [3.70]
Quintile	Name	VW	0.26 [3.71]	0.31 [4.44]	0.32 [4.58]	0.28 [4.01]	0.26 [3.79]	0.24 [3.48]
Quintile	Cap	VW	0.27 [3.89]	0.31 [4.58]	0.33 [4.78]	0.28 [4.02]	0.24 [3.47]	0.21 [3.02]
Decile	NYSE	VW	0.24 [2.47]	0.30 [3.10]	0.29 [2.91]	0.20 [2.02]	0.15 [1.57]	0.10 [1.00]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.22 [3.13]	0.28 [3.94]	0.28 [3.98]	0.25 [3.54]	0.21 [3.00]	0.18 [2.67]
Quintile	NYSE	EW	-0.08 [-1.41]					
Quintile	Name	VW	0.20 [2.84]	0.24 [3.47]	0.25 [3.58]	0.23 [3.29]	0.20 [2.93]	0.18 [2.71]
Quintile	Cap	VW	0.22 [3.13]	0.26 [3.85]	0.27 [4.00]	0.25 [3.61]	0.20 [2.92]	0.18 [2.64]
Decile	NYSE	VW	0.17 [1.74]	0.23 [2.34]	0.22 [2.19]	0.17 [1.72]	0.11 [1.13]	0.07 [0.75]

Table 3: Conditional sort on size and DFP

This table presents results for conditional double sorts on size and DFP. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on DFP. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high DFP and short stocks with low DFP. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	DFP Quintiles					DFP Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.66 [2.41]	0.89 [3.23]	0.97 [3.53]	0.96 [3.44]	0.75 [2.67]	0.10 [1.04]	0.12 [1.26]	0.09 [0.94]	0.06 [0.65]	0.05 [0.48]	0.03 [0.34]
	(2)	0.73 [2.73]	0.90 [3.58]	0.83 [3.28]	0.93 [3.76]	0.84 [3.35]	0.11 [1.50]	0.15 [1.99]	0.12 [1.65]	0.12 [1.52]	0.08 [1.04]	0.08 [1.02]
	(3)	0.82 [3.26]	0.76 [3.43]	0.84 [3.51]	0.82 [3.62]	0.91 [3.89]	0.10 [1.23]	0.15 [1.89]	0.15 [1.85]	0.11 [1.40]	0.14 [1.78]	0.12 [1.47]
	(4)	0.71 [3.17]	0.81 [3.84]	0.83 [3.73]	0.76 [3.58]	0.87 [4.06]	0.16 [2.13]	0.19 [2.61]	0.18 [2.45]	0.17 [2.28]	0.13 [1.75]	0.13 [1.69]
	(5)	0.45 [2.24]	0.62 [3.45]	0.60 [3.16]	0.67 [3.67]	0.82 [4.25]	0.37 [4.10]	0.40 [4.42]	0.41 [4.67]	0.34 [3.78]	0.32 [3.64]	0.27 [3.06]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	DFP Quintiles					DFP Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	386	386	388	386	383	36	33	32	32	35	
	(2)	106	106	105	106	105	59	61	59	60	60	
	(3)	75	75	75	75	75	107	107	103	105	106	
	(4)	63	63	63	63	63	227	235	226	232	225	
(5)	58	58	58	58	58	1448	2040	1875	2220	1464		

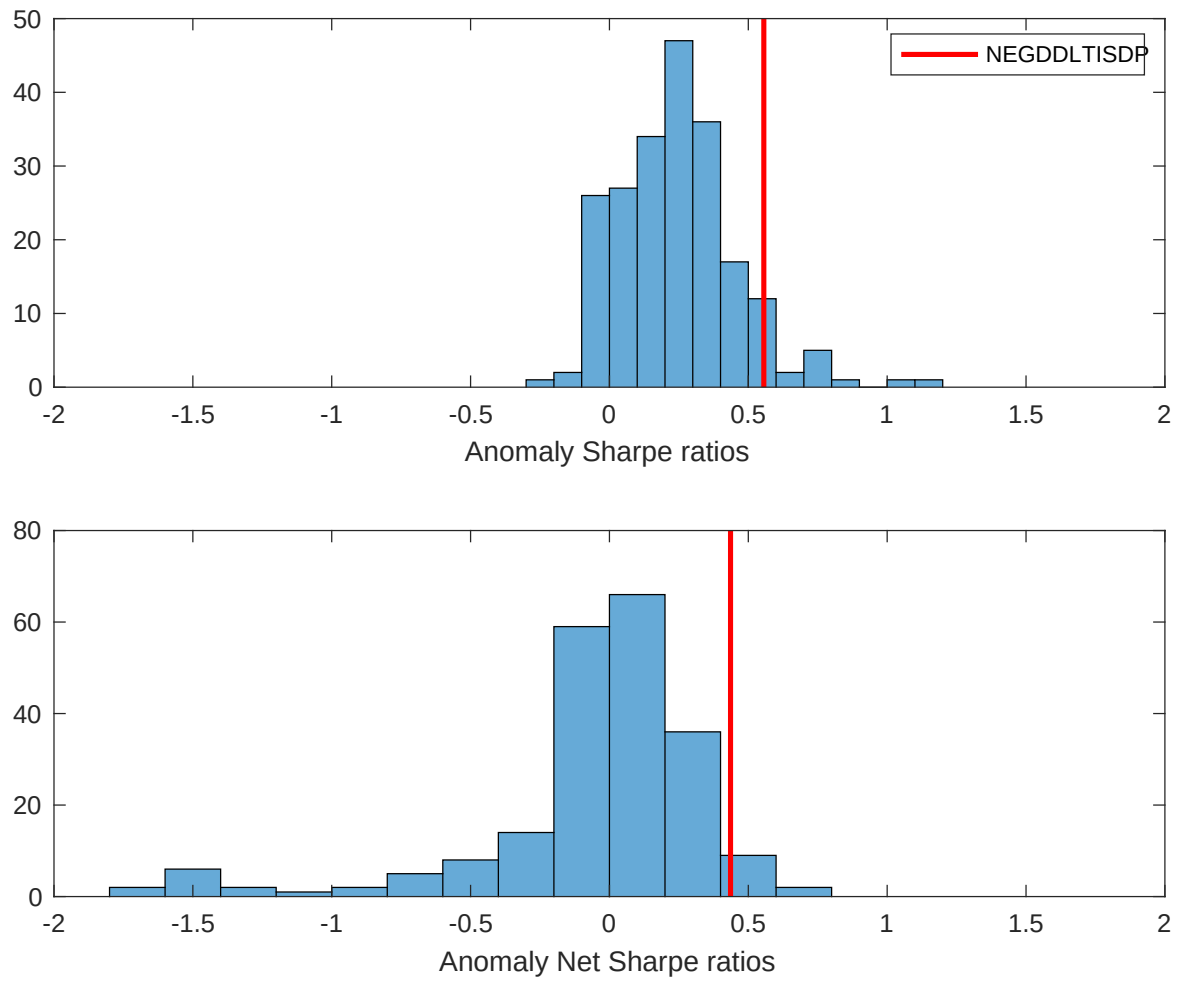


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the DFP with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

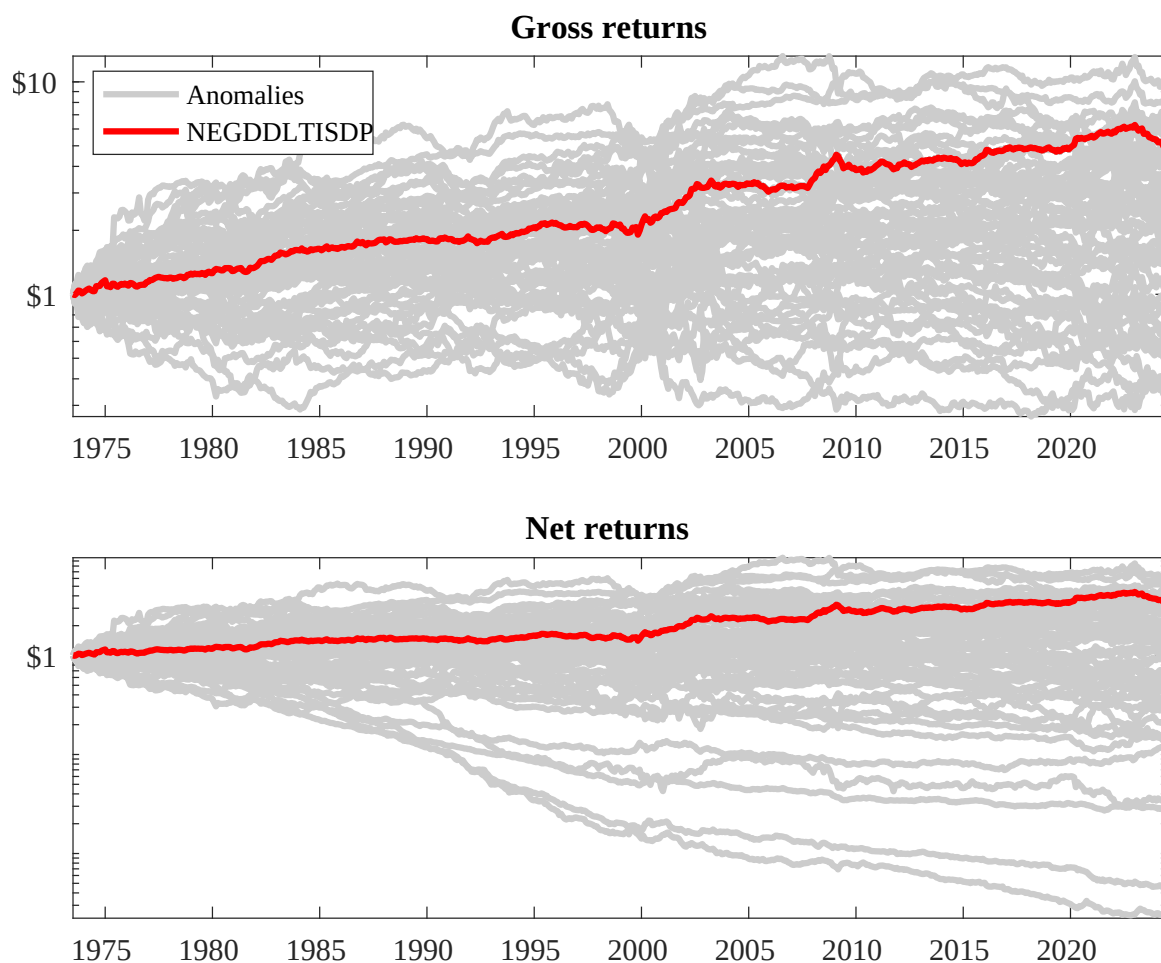
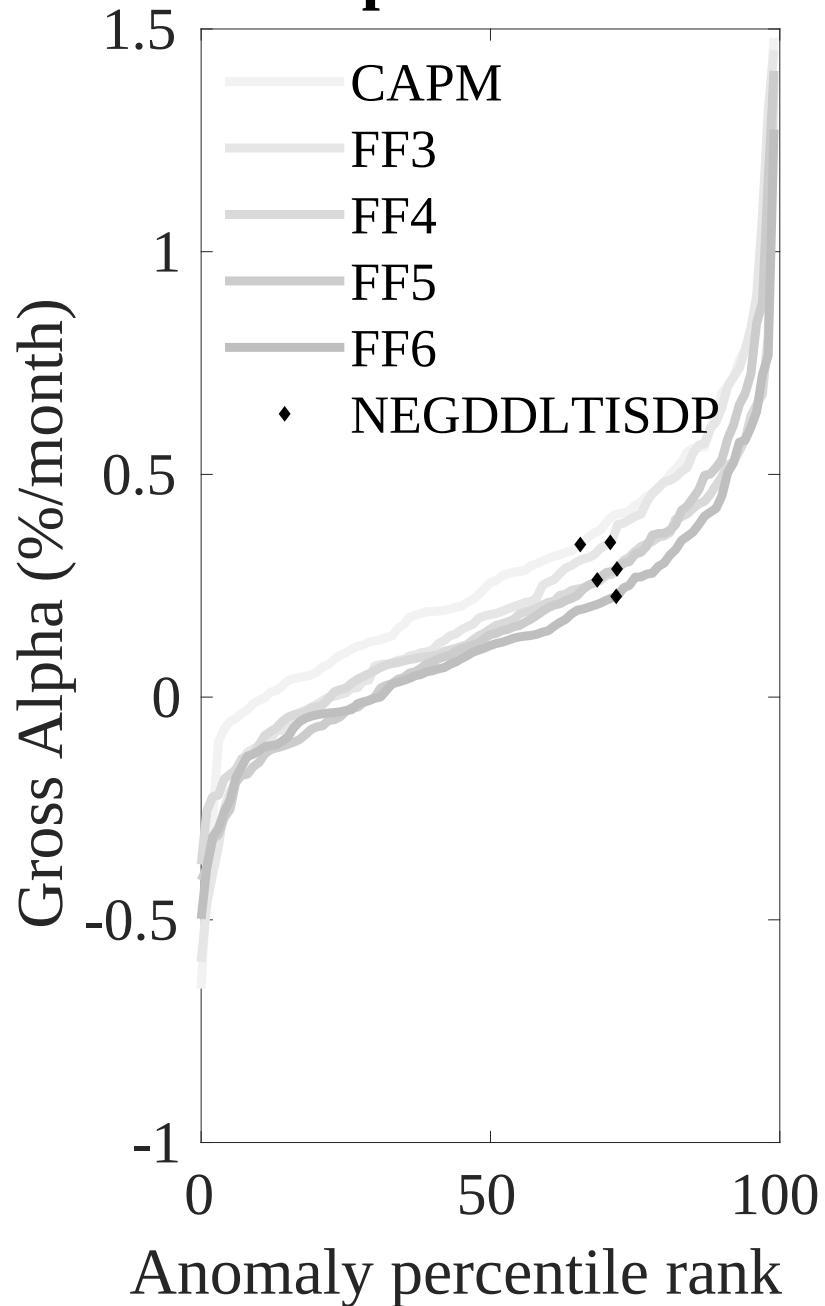


Figure 3: Dollar invested.

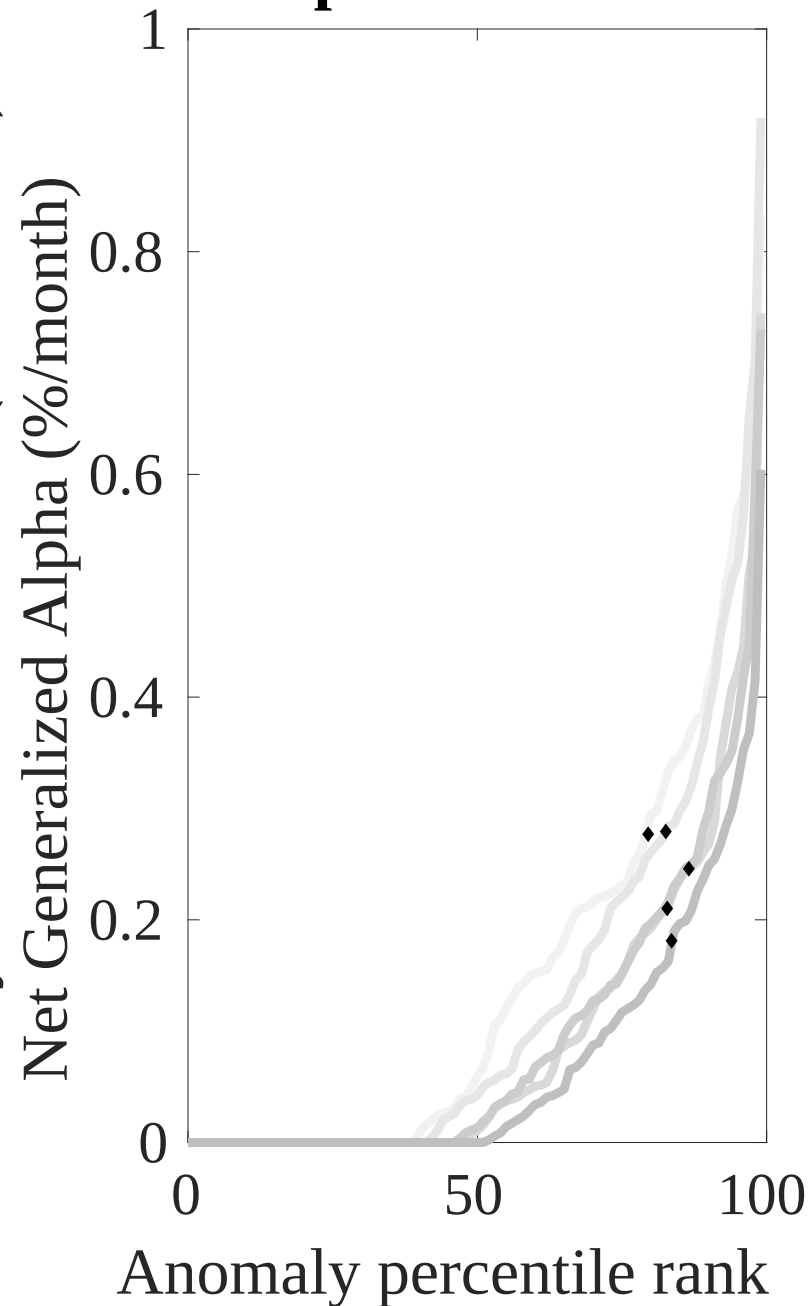
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the DFP trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



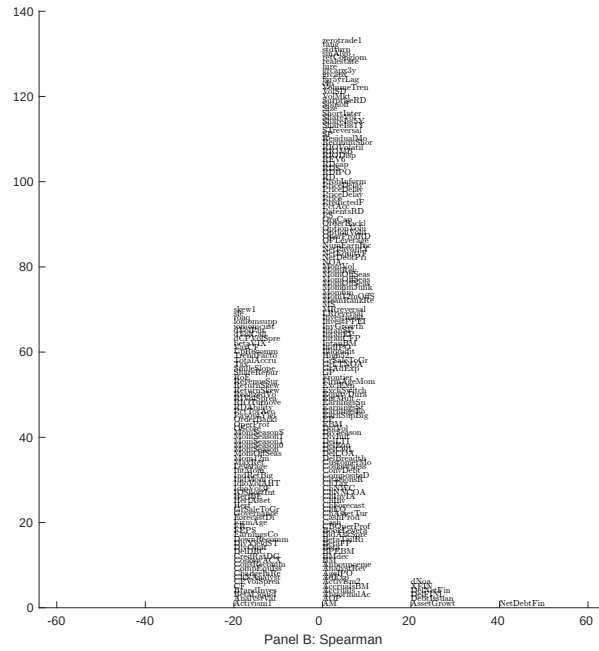
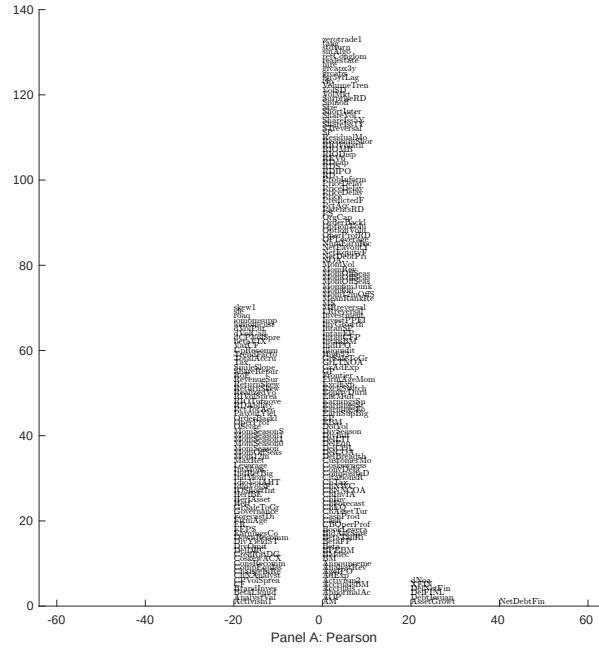


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with DFP. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

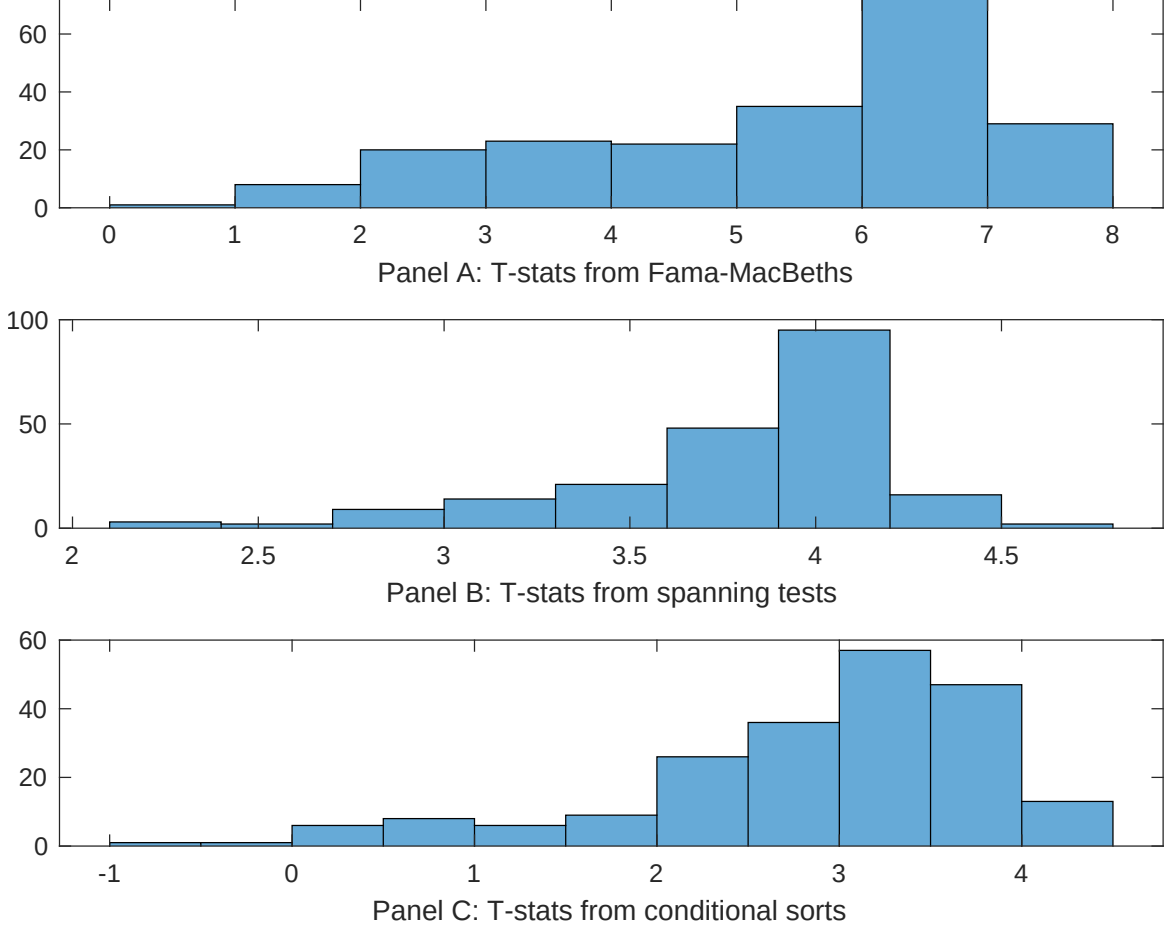


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of DFP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DFP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DFP}DFP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DFP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on DFP. Stocks are finally grouped into five DFP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DFP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on DFP, and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{DFP}DFP_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.39]	0.13 [5.36]	0.14 [5.72]	0.14 [5.82]	0.14 [5.31]	0.13 [5.37]	0.15 [5.65]
DFP	0.46 [2.32]	0.58 [2.70]	0.76 [3.65]	0.38 [1.79]	0.12 [5.37]	0.11 [5.59]	0.16 [0.62]
Anomaly 1	0.17 [9.28]						-0.58 [-1.26]
Anomaly 2		0.20 [8.70]					0.93 [1.38]
Anomaly 3			0.19 [6.23]				0.10 [1.74]
Anomaly 4				0.11 [8.47]			0.73 [3.52]
Anomaly 5					0.40 [6.88]		0.93 [1.68]
Anomaly 6						0.92 [6.05]	0.92 [0.66]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	1	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the DFP trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{DFP} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.19 [2.88]	0.20 [2.95]	0.20 [2.93]	0.21 [3.12]	0.22 [3.14]	0.22 [3.10]	0.18 [2.66]
Anomaly 1	18.20 [4.74]						9.43 [1.85]
Anomaly 2		20.55 [5.58]					12.21 [2.40]
Anomaly 3			11.88 [3.47]				8.87 [2.42]
Anomaly 4				0.78 [0.19]			-3.74 [-0.88]
Anomaly 5					1.83 [0.68]		1.44 [0.52]
Anomaly 6						0.28 [0.08]	-1.51 [-0.39]
mkt	-5.11 [-3.27]	-5.49 [-3.55]	-3.91 [-2.38]	-5.55 [-3.49]	-5.58 [-3.51]	-5.54 [-3.48]	-4.14 [-2.52]
smb	3.69 [1.54]	3.74 [1.58]	9.37 [3.58]	5.51 [2.28]	5.76 [2.38]	5.58 [2.31]	6.68 [2.41]
hml	-13.28 [-4.40]	-13.65 [-4.57]	-13.16 [-4.31]	-14.49 [-4.72]	-14.52 [-4.74]	-14.51 [-4.61]	-11.97 [-3.85]
rmw	-0.86 [-0.28]	-1.48 [-0.48]	-6.32 [-1.71]	0.78 [0.25]	1.05 [0.33]	0.79 [0.25]	-6.59 [-1.77]
cma	26.40 [5.57]	27.17 [5.86]	24.74 [4.84]	31.52 [4.70]	30.84 [5.91]	32.21 [5.65]	25.00 [3.56]
umd	4.42 [2.79]	4.67 [3.00]	5.91 [3.78]	6.02 [3.81]	5.81 [3.63]	5.98 [3.75]	4.20 [2.61]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	19	20	17	16	16	16	20

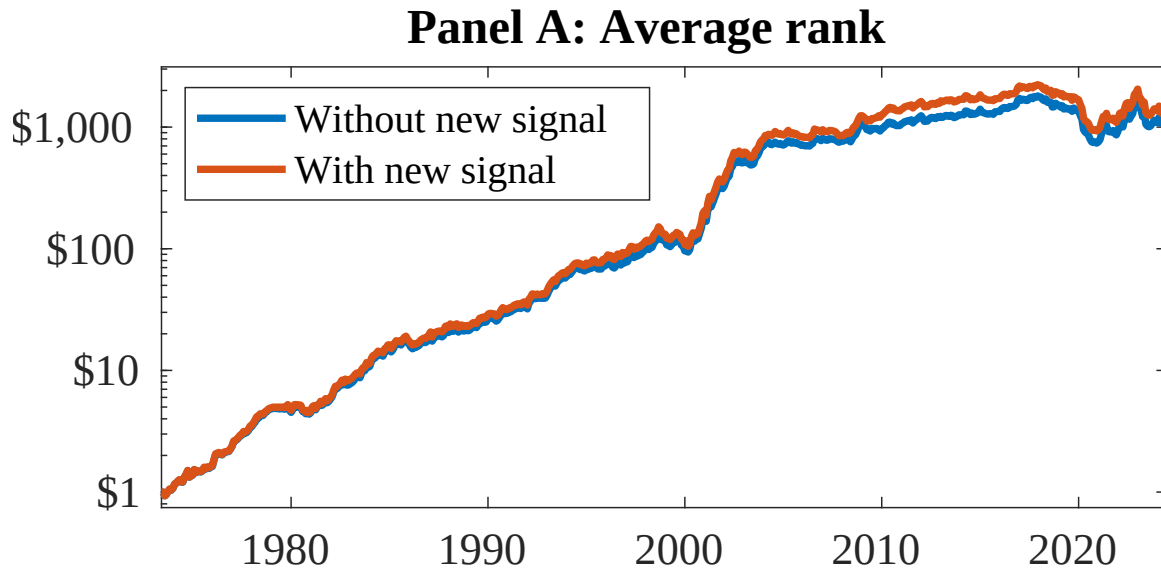


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as DFP. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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